(10) Publication No: **IN202541112613 A1**
(22) Date of Filing: 17-11-2025
(43) Publication Date: 10-04-2026
Journal No: 15/2026

(19) Intellectual Property Office, India

(12) INDIAN PATENT APPLICATION

(54) Title: **A ceiling fan-mounted hybrid thermoelectric cooling and dehumidification system and a method thereof**

(51) International Classification:
H10F 77/63, H01L 23/38, F02M 35/06, F25D 16/00 H02J 9/02

(21) Application No: **202541112613**

(31) Priority Document No: --

(32) Priority Date: --

(86) International Application No: --
Filing Date: --

(87) International Publication No: --

(61) Patent of Addition to Application No: --
Filing Date: --

(62) Divisional to Application No: --
Filing Date: --

(71) Name of Applicant(s):
Manipal Academy Of Higher Education,

(72) Name of Inventor(s):
1. Vaibhav Jain
2. Aaditya Krishna
3. Nathan Abey Ponnan
4. Ryan Bantu

(57) Abstract:

A CEILING FAN-MOUNTED HYBRID THERMOELECTRIC COOLING AND DEHUMIDIFICATION SYSTEM AND A METHOD THEREOF The current invention relates to a hybrid thermoelectric cooling and dehumidification system mounted on a ceiling fan, along with its associated method. The system combines a Peltier cooling module, a reusable desiccant dehumidifier and a phase change material (PCM) heat sink into one retrofit unit. Warm, moist air is pulled in, dehumidified and cooled prior to being released into the fan's downdraft, ensuring localised comfort. Functioning at around 120 W and generating under 55 dB of noise, the system does not need refrigerants, water or any external setup. The cyclic process involves phases of active cooling, heat storage and desiccant regeneration, all managed by a microcontroller. The invention provides eco-friendly, affordable and refrigerant-free cooling ideal for tropical environments. FIG. 1